
PART TWO

PROGRAMING THEORY

Why We Need Teaching Machines

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Current suggestions for improving education are familiar to everyone. We need more and better schools and colleges. We must pay salaries which will attract and hold good teachers. We should group students according to ability. We must bring textbooks and other materials up to date, particularly in science and mathematics. And so on. It is significant that all this can be done without knowing much about teaching or learning. Those who are most actively concerned with improving education seldom discuss what is happening when a student reads a book, writes a paper, listens to a lecture, or solves a problem, and their proposals are only indirectly designed to make these activities more productive. In short, there is a general neglect of education method. (Television is no exception, for it is only a way of amplifying and extending *old* methods, together with their shortcomings.)

It is true that the psychology of learning has so far not been very helpful in education. Its learning curves and its theories of learning have not yielded greatly improved classroom practices. But it is too early to conclude that nothing useful is to be learned about the behavior of teacher and student. No enterprise can improve itself very effectively without examining its basic processes. Fortunately, recent advances in the experimental analysis of behavior suggest that a true technology of education is feasible. Improved techniques are available to carry out the two basic assignments of education: constructing extensive repertoires of verbal and nonverbal behavior and generating that high probability of action which is said to show interest, enthusiasm, or a strong "desire to learn."

The processes clarified by an experimental analysis of be-

havior have, of course, always played a part in education, but they have been used with little understanding of their effects, wanted or unwanted. Whether by intention or necessity, teachers have been less given to teaching than to holding students responsible for learning. Methods are still basically aversive. The student looks, listens, and answers questions (and, incidentally, sometimes learns) as a gesture of avoidance or escape. A good teacher can cite exceptions, but it is a mistake to call them typical. The birch rod and cane are gone, but their place has been taken by equally effective punishments (criticism, possibly ridicule, failure) used in the same way: the student must learn, or else!

By-products of aversive control in education range from truancy, early drop-outs, and school-vandalism to inattention, "mental fatigue," forgetting, and apathy. It does not take a scientific analysis to trace these to their sources in educational practice. But more acceptable techniques have been hard to find. Erasmus tells of an English gentleman who tried to teach his son Greek and Latin without punishment. He taught the boy to use a bow and arrow and set up targets in the shape of Greek and Latin letters, rewarding each hit with a cherry. He also fed the boy letters cut from delicious biscuits. As a result, we may assume that the boy salivated slightly upon seeing a Greek or Latin text and that he was probably a better archer; but any effect on his knowledge of Greek and Latin is doubtful.

Current efforts to use rewards in education show the same indirection. Texts garnished with pictures in four colors, exciting episodes in a scientific film, interesting classroom activities—these will make a school interesting and even attractive (just as the boy probably liked his study of Greek and Latin), but to generate specific forms of behavior these things must be related to the student's behavior in special ways. Only then will they be truly rewarding or, technically speaking, "reinforcing."

We make a reinforcing event contingent on behavior when, for example, we design a piece of equipment in which a hungry rat or monkey or chimpanzee may press a lever and immediately obtain a bit of food. Such a piece of equipment gives us a powerful control over behavior. By scheduling reinforcements, we may maintain the behavior of pressing the lever in any given strength for long periods of time. By reinforcing special kinds of responses to the lever—for example, very light or very heavy presses or those made with one hand

or the other—we “shape” different forms or topographies of behavior. By reinforcing only when particular stimuli or classes of stimuli are present, we bring the behavior under the control of the environment. All these processes have been thoroughly investigated, and they have already yielded standard laboratory practices in manipulating complex forms of behavior for experimental purposes. They are obviously appropriate to educational design.

In approaching the problem of the educator we may begin by surveying available reinforcers. What positive reasons can we give the student for studying? We can point to the ultimate advantages of an education—to the ways of life which are open only to educated men—and the student himself may cite these to explain why he wants an education, but ultimate advantages are not contingent on behavior in ways which generate action. Many a student can testify to the result. No matter how much he may *want* to become a doctor or an engineer, say, he cannot force himself to read and remember the page of text in front of him at the moment. All notions of ultimate utility (as, for example, in economics) suffer from the same shortcoming: they do not specify effective contingencies of reinforcement.

The gap between behavior and a distant consequence is sometimes bridged by a series of “conditioned reinforcers.” In the laboratory experiment just described a delay of even a fraction of a second between the response to the lever and the appearance of food may reduce the effectiveness of the food by a measurable amount. It is standard practice to let the movement of a lever produce some visual stimulus, such as a change in the illumination in the apparatus, which is then followed by food. In this way the change in illumination becomes a conditioned reinforcer which can be made immediately contingent on the response. The marks, grades, and diplomas of education are conditioned reinforcers designed to bring ultimate consequences closer to the behavior reinforced. Like prizes and medals, they represent the approval of teachers, parents, and others, and they show competitive superiority, but they are mainly effective because they signalize progress through the system—toward some ultimate advantage of, or at least freedom from, education. To this extent they bridge the gap between behavior and its remote consequences; but they are still not contingent on behavior in a very effective way.

Progressive education tried to replace the birch rod, and at

the same time avoid the artificiality of grades and prizes, by bringing the reinforcers of everyday life into the schools. Such natural contingencies have a kind of guaranteed effectiveness. But a school is only a small part of the student's world, and no matter how real it may seem, it cannot provide natural reinforcing consequences for all the kinds of behavior which education is to set up. The goals of progressive education were shifted to conform to this limitation, and many worthwhile assignments were simply abandoned.

Fortunately, we can solve the problem of education without discovering or inventing additional reinforcers. We merely need to make better use of those we have. Human behavior is distinguished by the fact that it is affected by small consequences. Describing something with the right word is often reinforcing. So is the clarification of a temporary puzzlement, or the solution of a complex problem, or simply the opportunity to move forward after completing one stage of an activity. We need not stop to explain *why* these things are reinforcing. It is enough that, when properly contingent upon behavior, they provide the control we need for successful educational design. Proper contingencies of reinforcement, however, are not always easily arranged. A modern laboratory for the study of behavior contains elaborate equipment designed to control the environment of individual organisms during many hours or days of continuous study. The required conditions and changes in conditions cannot be arranged by hand, not only because the experimenter does not have the time and energy, but because many contingencies are too subtle and precise to be arranged without instrumental help. The same problem arises in education.

Consider, for example, the temporal patterning of behavior called "rhythm." Behavior is often effective only if properly timed. Individual differences in timing, ranging from the most awkward to the most skillful performances, affect choice of career and of artistic interests and participation in sports and crafts. Presumably a "sense of rhythm" is worth teaching, yet practically nothing is now done to arrange the necessary contingencies of reinforcement. The skilled typist, tennis player, lathe operator, or musician is, of course, under the influence of reinforcing mechanisms which generate subtle timing, but many people never reach the point at which these natural contingencies can take over.

A relatively simple device supplies the necessary contingencies. The student taps a rhythmic pattern in unison with

the device. "Unison" is specified very loosely at first (the student can be a little early or late at each tap) but the specifications are slowly sharpened. The process is repeated for various speeds and patterns. In another arrangement, the student echoes rhythmic patterns sounded by the machine, though not in unison, and again the specifications for an accurate reproduction are progressively sharpened. Rhythmic patterns can also be brought under the control of a printed score.

Another kind of teaching machine generates sensitivity to properties of the environment. We call an effective person "discriminating." He can tell the difference between the colors, shapes, and sizes of objects, he can identify three-dimensional forms seen from different aspects, he can find patterns concealed in other patterns, he can identify pitches, intervals, and musical themes and distinguish between different tempos and rhythms—and all of this in an almost infinite variety. Subtle discriminations of this sort are as important in science and industry and in everyday life as in identifying the school of a painter or the period of a composer.

The ability to make a given kind of discrimination can be taught. A pigeon, for example, can be *made* sensitive to the color, shape, and size of objects, to pitches, and rhythms, and so on—simply by reinforcing it when it responds in some arbitrary way to one set of stimuli and extinguishing responses to all others. The same kinds of contingencies of reinforcement are responsible for human discriminative behavior. *The remarkable fact is that they are quite rare in the environment of the average child.* True, children are encouraged to play with objects of different sizes, shapes, and colors, and are given a passing acquaintance with musical patterns; but they are seldom exposed to the precise contingencies needed to build subtle discriminations. It is not surprising that most of them move into adulthood with largely undeveloped "abilities."

The number of reinforcements required to build discriminative behavior in the population as a whole is far beyond the capacity of teachers. Too many teachers would be needed, and many contingencies are too subtle to be mediated by even the most skillful. *Yet relatively simple machines will suffice.* The apparatus shown in Figure 1 is adapted from research on lower organisms. It teaches an organism to discriminate selected properties of stimuli while "matching to sample." Pictures or words are projected on translucent windows which respond to a touch by closing circuits. A child can be made

to "look at the sample" by reinforcing him for pressing the top window. An adequate reinforcement for this response is simply the appearance of material in the lower windows, from which a choice is to be made.

The child identifies the material which corresponds to the sample in some prescribed way by pressing one of the lower windows, and he is then reinforced again—possibly simply be-

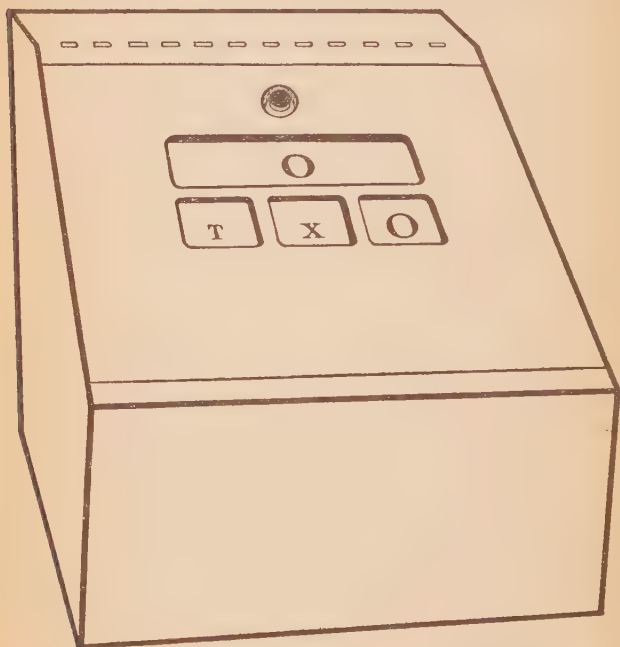


Fig. 1. A machine to teach the matching of colors, shapes, sizes, as well as correspondences between pictures and words, words and other words, and so on.

cause a new set of materials now appears on the windows. If he presses the wrong window, all three choices disappear until the top window has been pressed again—which means until he has again looked at the sample. Many other arrangements of responses and reinforcements are, of course, possible. In an auditory version, the child listens to a sample pattern of tones and then explores other samples to find a match.

If devices similar to these were generally available in our

nursery schools and kindergartens, our children would be far more skillful in dealing with their environments. They would be more productive in their work, more sensitive to art and music, better at sports, and so on. They would lead more effective lives. We cannot assert all this with complete confidence on the present evidence, but there is no doubt whatsoever *that the conditions needed to produce such a state of affairs are now lacking*. In the light of what we know about differential contingencies of reinforcement, the world of the young child is shamefully impoverished. And only machines will remedy this, for the required frequency and subtlety of reinforcement cannot otherwise be arranged.

The teacher is, of course, at a disadvantage in teaching skilled and discriminative behavior, because such instruction is largely nonverbal. It may be that the methods of the classroom, in which the teacher is said to "communicate" with the student, to "impart information," and to build "verbal abilities," are better adapted to standard subject matters, the learning of which is usually regarded as more than the acquisition of forms of behavior or of environmental control. Yet a second look may be worth while. Traditional characterizations of verbal behavior raise almost insuperable problems for the teacher, and a more rigorous analysis suggests another possibility. We can define terms like "information," "knowledge," and "verbal ability" by reference to the behavior from which we infer their presence. *We may then teach the behavior directly*. Instead of "transmitting information to the student" we may simply set up the behavior which is taken as a sign that he possesses information. Instead of teaching a "knowledge of French" we may teach the behavior from which we infer such knowledge. Instead of teaching "an ability to read" we may set up the behavioral repertoire which distinguishes the child who knows how to read from one who does not.

To take the last example, a child reads or "shows that he knows how to read" by exhibiting a behavioral repertoire of great complexity. He finds a letter or word in a list on demand; he reads aloud; he finds or identifies objects described in a text; he rephrases sentences; he obeys written instructions; he behaves appropriately to described situations; he reacts emotionally to described events; and so on, in a long list. He does none of this before learning to read and all of it afterwards. To bring about such a change is an extensive assignment, and it is tempting to try to circumvent it by teaching something called "an ability to read" from which all these

specific behaviors will flow. But this has never actually been done. "Teaching reading" is always directed toward setting up specific items in such a repertoire.

It is true that parts of the repertoire are not independent. A student may acquire some kinds of responses more readily for having acquired others, and he may for a time use some in place of others (for example, he may follow written directions not by responding directly to a text but by following his own spoken instructions as he reads the text aloud). In the long run all parts of the repertoire tend to be filled in, not because the student is rounding out an ability to read, but simply because all parts are in their several ways useful. They all continue to be reinforced by the world at large after the explicit teaching of reading has ceased.

Viewed in this way, reading can also be most effectively taught with instrumental help. A pupil can learn to distinguish among letters and groups of letters in an alphabet simply as visual patterns in using the device and procedures just described. He can be taught to identify arbitrary correspondences (for example, between capitals and lower-case letters, or between handwritten and printed letters) in a more complex type of stimulus control which is within reach of the same device. With a phonographic attachment, correspondences between printed letters and sounds, between sounds and letters, between words and sounds, between sounds and printed words, and so on, can be set up. (The student could be taught all of this without pronouncing a word, and it is possible that he would learn good pronunciation more quickly if he had first done so.)

The same device can teach correspondences between words and the properties of objects. The pupil selects a printed or spoken word which corresponds in the language to, say, a pictured object or another printed or spoken word. These semantic correspondences differ in important respects from formal matches, but the same processes of programing and reinforcement can—indeed, must—be used. Traditional ways of teaching reading establish all these repertoires, but they do so indirectly and, alas, inefficiently. In "building a child's need to read," in motivating "his mental readiness," in "sharing information," and so on, the teacher arranges, sometimes almost surreptitiously, many of the contingencies just listed, and these are responsible for whatever is learned. An explicit treatment clarifies the program, suggests effective procedures, and guarantees a coverage which is often lacking with tradi-

tional methods. Much of what is called reading has not been covered, of course, but it may not need to be taught, for once these basic repertoires have been established, the child begins to receive automatic reinforcement in responding to textual material.

The same need for a behavioral definition arises in teaching other verbal skills (for example, a second language) as well as the traditional subjects of education. In advancing to that level, however, we must transcend a limitation of the device in Figure 1. The student can *select* a response without being able to speak or write, but we want him to learn to *emit* the response, since this is the kind of behavior which he will later find most useful. The emission of verbal behavior is taught by another kind of machine. A frame of textual material appearing in the square opening is incomplete: in place of certain letters or figures there are holes. Letters or figures can be made to appear in these holes by moving sliders (a keyboard would be an obvious improvement). When the material has been completed, the student checks his response by turning a crank. The machine senses the settings of the sliders and, if they are correct, moves a new frame of material into place, the sliders returning to their home position. If the response is wrong, the sliders return home, and a second setting must be made.

The machine can tell the student he is wrong without telling him what is right. This is an advantage, but it is relatively costly. Moreover, correct behavior is rather rigidly specified. Such a machine is probably suitable only for the lower grades. A simpler and cheaper procedure, with greater flexibility, is to allow the student to compare his written response with a revealed text. The device shown in Figure 2 uses this principle. It is suitable for verbal instruction beyond the lower primary grades—that is, through junior high school, high school, and college, and in industrial and professional education. Programed material is stored on fan-folded paper tapes. One frame of material, the size of which may be varied with the nature of the material, is exposed at a time. The student writes on a separate paper strip. He cannot look at unauthorized parts of the material without recording the fact that he has done so, because when the machine has been loaded and closed, it can be opened only by punching the strip of paper.

The student sees printed material in the large window at the left. This may be a sentence to be completed, a question to be answered, or a problem to be solved. He writes his response on an uncovered portion of a paper strip at the right.

He then moves a slider which covers the response he has written with a transparent mask and uncovers additional material in the larger opening. This may tell him that his response is wrong without telling him what is right. For example, it may list a few of the commonest errors. If the response he wrote is among them, he can try again on a newly uncovered portion of the paper strip. A further operation of the machine covers his second attempt and uncovers the correct response.

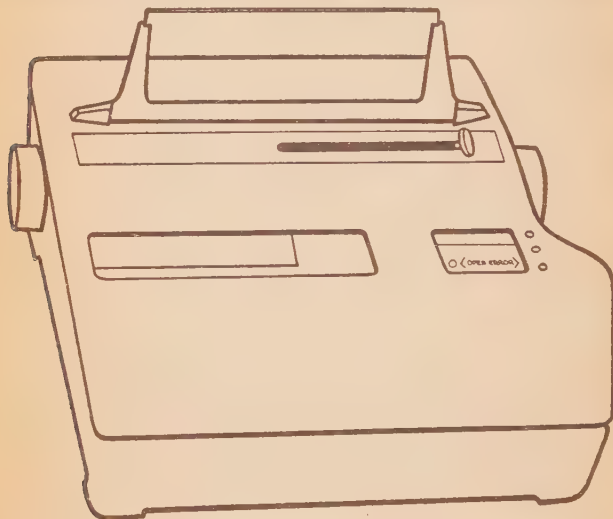


Fig. 2. A machine to teach "verbal knowledge."

The student records a wrong response by punching a hole alongside it, leaving a record for the instructor who may wish to review a student's performance, and operating a counter which becomes visible at the end of the set. Then the student records the number of mistakes he has made and may compare it with a par score for the set.

Exploratory research in schools and colleges indicates that what is now taught by teacher, textbook, lecture, or film can be taught in half the time with half the effort by a machine of this general type.¹ One has only to see students at work to

¹ Under the direction of Allen Calvin of Hollins College, an eighth-grade class in the Roanoke School System completed all the work of a ninth-grade class in algebra in one term. Test scores were comparable with a normal ninth-grade performance, and a test nine months later showed a retention of at least 90 per cent of the material learned.

understand why this is a conservative estimate. The student remains active. If he stops, the program stops (in marked contrast with classroom practice and educational television); but there is no compulsion, for he is not inclined to stop. Immediate and frequent reinforcement sustains a lively interest. (The interest, incidentally, outlasts any effect of novelty. Novelty may be relevant to interest, but the material in the machine is always novel.) Where current instructional procedures are highly efficient, the gain may not be so great. In one experiment² involving industrial education there was approximately a 25 per cent saving in the time required for instruction, something of the order of a 10 per cent increase in retention, and about 90 per cent of the students preferred to study by machine. In general, the student generally likes what he is doing; he makes no effort to escape—for example, by letting his attention wander. He need not force himself to work and is usually free of the feeling of effort generated by aversive control. He has no reason to be anxious about impending examinations, for none are required. Both he and his instructor know where he stands at all times.

No less important in explaining the success of teaching machines is the fact that each student is free to proceed at his own rate. Holding students together for instructional purposes in a class is probably the greatest source of inefficiency in education. Some efforts to mechanize instruction have missed this point. A language laboratory controlled from a central console presupposes a group of students advancing at about the same rate, even though some choice of material is permitted. Television in education has made the same mistake on a colossal scale. A class of twenty or thirty students moving at the same pace is inefficient enough, but what must we say of all the students in half a dozen states marching in a similar lock step?

In trying to teach more than one student at once we harm both fast and slow learners. The plight of the good student has been recognized, but the slow learner suffers more disastrous consequences. The effect of pressure to move beyond one's natural speed is cumulative. The student who has not fully mastered a first lesson is less able to master a second. His ultimate failure may greatly exaggerate his shortcoming; a

² More recent results with the same material improved in the light of the earlier experiment were reported by J. L. Hughes and W. J. McNamara at the Annual Meeting of the American Psychological Association in New York, September, 1961. Their work concerned the use of programed texts in industrial education.

small difference in speed has grown to an immense difference in comprehension. Some of those most active in improving education have been tempted to dismiss slow students impatiently as a waste of time, but it is quite possible that many of them are capable of substantial, even extraordinary, achievements if permitted to move at their own pace. Many distinguished scientists, for example, have appeared to think slowly.

One advantage of individual instruction is that the student is able to follow a program without breaks or omissions. A member of a class moving at approximately the same rate cannot always make up for absences, and limitations of contact time between student and teacher make it necessary to abbreviate material to the point at which substantial gaps are inevitable. Working on a machine, the student can always take up where he left off or, if he wishes, review earlier work after a longer absence. The coherence of the program helps to maximize the student's success, for by thoroughly mastering one step he is optimally prepared for the next. Many years ago, in their *Elementary Principles of Education*,³ Thorndike and Gates considered the possibility of a book "so arranged that only to him who had done what was directed on page one would page two become visible, and so on." With such a book, they felt, "much that now requires personal instruction could be managed by print." The teaching machine is, of course, such a book.

In summary, then, machine teaching is unusually efficient because (1) the student is frequently and immediately reinforced, (2) he is free to move at his natural rate, and (3) he follows a coherent sequence. These are the more obvious advantages, and they may well explain current successes. But there are more promising possibilities: the conditions arranged by a good teaching machine make it possible to apply to education what we have learned from laboratory research and to extend our knowledge through rigorous experiments in schools and colleges.

The conceptions of the learning process which underlie classroom practices have long been out of date. For example, teachers and textbooks are said to "impart information." They expose the student to verbal and nonverbal material and call attention to particular features of it, and in so doing they are said to "tell the student something." In spite of discouraging

³ Thorndike, Edward & Gates, Arthur. *Elementary Principles of Education*. New York: B. Macmillan Co., 1929.

evidence to the contrary, it is still supposed that if you tell a student something, he then knows it. In this scheme, teaching is the transmission of information, a notion which, through a false analogy, has acquired undue prestige from communication engineering. Something is undoubtedly transmitted by teacher to student, for if communication is interrupted, instruction ceases; but the teacher is not merely a source from which knowledge flows into the student. We cannot necessarily improve instruction by altering the conditions of transmission—as, for example, by changing to a different sensory modality. This is a mistake made by some so-called teaching machines which, accepting our failure to teach reading, have tried to restore communication by using recorded speech. The student no longer pores over a book, as in the traditional portraits; he stares into space with earphones on his head. For the same reasons improvements in the coding of information may not be immediately relevant.

The student is more than a receiver of information. He must take some kind of action. The traditional view is that he must “associate.” The stream of information flowing from teacher to student contains pairs of items which, being close together or otherwise related, become connected in the student’s mind. This is the old doctrine of the association of ideas, now strengthened by a scientific, if uncritical, appeal to conditioned reflexes: two things occurring together in experience somehow become connected so that one of them later reminds the student of the other. The teacher has little control over the process except to make sure that things occur together often and that the student pays attention to them—for example, by making the experiences vivid or, as we say, memorable. Some devices called teaching machines are simply ways of presenting things together in ways which attract attention. The student listens to recorded speech, for example, while looking at pictures. The theory is that he will associate these auditory and visual presentations.

But the action demanded of the student is not some sort of mental association of contiguous experiences. It is more objective and, fortunately, more controllable than that. To acquire behavior, *the student must engage in behavior*. This has long been known. The principle is implied in any philosophy of “learning by doing.” But it is not enough simply to acknowledge its validity. Teaching machines provide the conditions needed to apply the principle effectively.

Only in the early stages of education are we mainly inter-

ested in establishing *forms* of behavior. In the verbal field, for example, we teach a child to speak, eventually with acceptable accent and pronunciation, and later to write and spell. After that, topography of behavior is assumed; the student can speak and write and must now learn to do so appropriately—that is, he must speak or write in given ways under given circumstances. How he comes to do so is widely misunderstood. Education usually begins by establishing so-called formal repertoires. The young child is taught to “echo” verbal behavior in the sense of repeating verbal stimuli with reasonable accuracy. A little later he is taught to read—to emit verbal behavior under the control of textual stimuli. These and other formal repertoires are used in later stages of instruction to evoke new responses without “shaping” them.

In an important case of what we call instruction, control is simply transferred from so-called formal to thematic stimuli. When a student learns to memorize a poem, for example, it is clearly inadequate to say that by reading the poem he presents to himself its various parts contiguously and then associates them. He does not simply read the poem again and again until he knows it. (It is possible that he could learn the poem in that way.) Something else must be done, as anyone knows who has memorized a poem from the text. The student must make tentative responses while looking away from the text. He must glance at the text from time to time to provide fragmentary help in emitting a partially learned response. If a recalled passage makes sense, it may provide its own automatic confirmation, but if the passage is fragmentary or obscure, the student must confirm the correctness of an emitted response by referring to the text after he has emitted it.

A teaching machine facilitates this process. It presents the poem line by line and asks the student to read it. The text is then “vanished”—that is, it becomes less and less clear or less and less complete in subsequent presentations. Other stimuli (arising from the student’s own behavior in this case) take over. In one procedure a few unimportant letters are omitted in the first presentation. The student reads the line without their help and indicates his success by writing down the omitted letters, which are confirmed by the machine. More of the line is missing when it again appears, but because he has recently responded to a fuller text, the student can nevertheless read it correctly. Eventually, no textual stimulus remains, and he can “recite” the poem.

(If the reader wishes to try this method on a friend or mem-

ber of his family without a machine, he may do so by writing the poem on a chalk board in a clear hand, omitting a few unimportant letters. He should ask his subject to read the poem aloud but to make no effort to memorize it. He should then erase another selection of letters. He will have to guess at how far he can go without interfering with his subject's success on the next reading, but under controlled conditions this could be determined for the average student quite accurately. Again the subject reads the poem aloud, making no effort to memorize, though he may have to make some effort to recall. Other letters are then erased and the process repeated. For a dozen lines of average material, four or five readings should suffice to eliminate the text altogether. The poem can still be "read.")

Memorized verbal behavior is a valuable form of knowledge which has played an important role in classical education. There are other, and generally more useful, forms in which the same processes are involved. Consider, for example, a labeled picture. To say that such an instructional device "tells the student the name of the pictured object" is highly elliptical—and dangerous if we are trying to understand the processes involved. Simply showing a student a labeled picture is no more effective than letting him read a poem. He must take some sort of action. As a formal stimulus, the label evokes a verbal response, not in this case in the presence of other verbal behavior on the part of the student, but in the presence of the picture. The control of the response is to pass from the label to the picture; the student is to give the name of the pictured object without reading it.

The steps taken in teaching with labeled pictures can also be arranged particularly well with a machine. Suppose we are teaching medical-school anatomy at the textbook level. Certain labeled charts represent what is to be learned in the sense that the student will eventually (1) give the names of indicated parts and describe relations among them and (2) be able to point to, draw, or construct models of parts, or relations among them, given their names. To teach the first of these, we induce the student to describe relations among the parts shown on a fully labeled chart. One effect of this is that he executes the verbal behavior at issue—he writes the names of the parts. More important, he does this while, or just after, looking at corresponding pictured details. He will be able to write the names again while looking at a chart which shows only incomplete names, possibly only initial letters. Finally, he will

be able to supply the complete names of parts identified only by number on still another chart. His verbal responses have passed from the control of textual stimuli to that of pictured anatomical details. Eventually, as he studies a cadaver, the control will pass to the actual anatomy of the human body. In this sense he then "knows the names of the parts of the body and can describe relations among them."

(The device shown in Figure 2 is designed to skip one or two steps in "vanishing" textual stimuli. A fully labeled chart may be followed by a merely numbered one. The student writes the name corresponding to a number in the first space. If he cannot do this, he operates the machine to uncover, not merely some indication that he is right or wrong, but additional help—say, a few letters of the correct response.)

Learning a poem or the names of pictured objects is a relatively straightforward task. More complex forms of knowledge require other procedures. At an early point, the main problem becomes that of analyzing knowledge. Traditionally, for example, something called a "knowledge of French" is said to permit the student who possesses it to do many things. One who possesses it can (1) repeat a French phrase with a good accent, (2) read a French text in all the senses of reading listed above, (3) take dictation in French, (4) find a word spoken in French on a printed list, (5) obey instructions spoken in French, (6) comment in French upon objects or events, (7) give orders in French, and so on. If he also "knows English," he can give the English equivalents of French words or phrases or the French equivalents of English words or phrases.

The concept of "a knowledge of French" offers very little help to the would-be teacher. As in the case of reading, we must turn to the behavioral repertoires themselves, for these are all that have ever been taught when education has been effective. The definition of a subject matter in such terms may be extraordinarily difficult. Students who are "competent in first-year college physics," for example, obviously differ from those who are not—but in what way? Even a tentative answer to that question should clarify the problem of teaching physics. It may well do more. In the not-too-distant future much more general issues in epistemology may be approached from the same direction. It is possible that we shall fully understand the nature of knowledge only after having solved the practical problems of imparting it.

Until we can define subject matters more accurately and

until we have improved our techniques of building verbal repertoires, writing programs for teaching machines will remain something of an art. This is not wholly satisfactory, but there is some consolation in the fact that an impeccable authority on the excellence of a program is available. The student himself can tell the programmer where he has failed. By analyzing the errors made by even a small number of students in a pilot study, it is usually possible to work a great improvement in an early version of a program. (The machine shown in Figure 2 is designed to supply the necessary feedback to the programmer in convenient form. When a student punches an error, he marks the back of the printed material, which eventually carries an item-by-item record of the success or failure of the programmer. This is obviously valuable during the experimental stages of programing, but it will also be desirable when machines are widely used in schools and colleges, since publishers can then periodically call in programs to be studied and improved by their authors. The information supplied might be compared to a record showing the percentage of students who have misunderstood each sentence in a text.)

The teaching machine shown in Figure 2 falls far short of the "electronic classrooms" often visualized for the schools and colleges of the future. Many of these, often incorporating small computers, are based on misunderstandings of the learning process. They are designed to duplicate current classroom conditions. When instruction is badly programed, a student often goes astray, and a teacher must come to his rescue. His mistakes must be analyzed and corrected. This may give the impression that instruction is largely a matter of correcting errors. If this were the case, an effective machine would, indeed, have to follow the student into many unprofitable paths and take remedial action. But under proper programing nothing of this sort is required. It is true that a relatively important function of the teacher will be to follow the progress of each student and to suggest collateral material which may be of interest, as well as to outline further studies, to recommend changes to programs of different levels of difficulty, and so on, and to this extent a student's course of study will show "branching." But changes in level of difficulty or in the character of the subject need not be frequent and can be made as the student moves from one set of material to another.

Teaching machines based on the principle of "multiple choice" also often show a misunderstanding of the learning process. When multiple-choice apparatuses were first used,

the organism was left to proceed by "trial and error." The term does not refer to a behavioral process but simply to the fact that contingencies of reinforcement were left to chance: some responses happened to be successful and others not. Learning was not facilitated or accelerated by procedures which increased the probability of successful responses. The results, like those of much classroom instruction, suggested that errors were essential to the learning process. But when material is carefully programed, both subhuman and human subjects can learn while making few errors or even none at all. Recent research by Herbert S. Terrace,⁴ for example, has shown that a pigeon can learn to discriminate colors practically without making mistakes. The control exerted by color may be passed, *via* a vanishing technique, to more difficult properties of stimuli—again without error. Of course we learn something from our mistakes—for one thing, we learn not to make them again—but we *acquire* behavior in other ways.

The teaching machines of S. L. Pressey,⁵ the first psychologist to see the "coming industrial revolution in education," were mechanical versions of self-scoring test forms, which Pressey and his students also pioneered. They were not designed for programed instruction in the present sense. The student was presumed to have studied a subject before coming to the machine. By testing himself, he consolidated what he had partially learned. For this purpose a device which evaluated the student's selection from an array of multiple-choice items was appropriate. For the same purpose multiple-choice material can, of course, be used in all the machines described above. But several advantages of programed instruction are lost when such material is used in straightforward instruction.

In the first place, the student should *construct* rather than *select* a response, since this is the behavior he will later find useful. Secondly, he should advance to the level of being able to emit a response rather than merely recognize a given response as correct. This represents a much more considerable achievement, as the difference between the sizes of reading and writing vocabularies in a foreign language demonstrates.

⁴ Terrace, Herbert S. *Discrimination Learning With and Without Errors* (unpublished Ph.D. Dissertation, Department of Psychology, Harvard University, 1961).

⁵ Pressey, S. L. A simple apparatus which gives tests and scores—and teaches. *School and Society*, 1926, 23, 373–376. (This article and other articles concerning teaching machines by S. L. Pressey are included in Lumsdaine, A. A. & Glaser, Robert (eds.), *Teaching Machines and Programmed Learning: A Source Book* (Washington, D.C.: National Education Association, 1960).

Thirdly, and more important, multiple-choice material violates a basic principle of good programing by inducing the student to engage in erroneous behavior. Those who have written multiple-choice tests know how much time, energy, and ingenuity are needed to construct plausible wrong answers. (They must be plausible or the test will be of little value.) In a multiple-choice *test*, they may do no harm, since a student who has already learned the right answer may reject wrong answers with ease and possibly with no undesirable side effects. The student who is *learning*, however, can scarcely avoid trouble. Traces of erroneous responses survive in spite of the correction of errors or the confirmation of a right answer. In multiple-choice material designed to teach "literary appreciation," for example, the student is asked to consider three or four plausible paraphrases of a passage in a poem and to identify the most acceptable. But as the student reads and considers unacceptable paraphrases, the very processes which the poet himself used in making his poem effective are at work to destroy it. Neither the vigorous correction of wrong choices nor the confirmation of a right choice will free the student of the verbal and nonverbal associations thus generated.

Scientific subjects offer more specific examples. Consider an item such as the following, which might be part of a course in high school physics:

As the pressure of a gas increases, volume decreases. This is because:

- (a) *the space between the molecules grows smaller*
- (b) *the molecules are flattened*
- (c) *etc. . . .*

Unless the student is as industrious and as ingenious as the multiple-choice programmer, it will probably not have occurred to him that molecules may be flattened as a gas is compressed (within the limits under consideration). If he chooses item (b) and is corrected by the machine, we may say that he "has learned that it is wrong," but this does not mean that the sentence will never occur to him again. And if he is unlucky enough to select the right answer first, his reading of the plausible but erroneous answer will be corrected only "by implication"—an equally vague and presumably less effective process. In either case, he may later find himself recalling that "somewhere he has read that molecules are flattened when a gas is compressed." And, of course, somewhere he has.

Multiple-choice techniques are appropriate when the student is to learn to compare and choose. In forming a discrimination (as with the device shown in Figure 1), an organism must be exposed to at least two stimuli, one of which may be said to be wrong. Similarly, in learning to "troubleshoot" equipment there may be several almost equally plausible ways of correcting a malfunction. Games offer other examples. A given hand at bridge may justify several bids or plays, no one of which is wholly right and all the others wrong. In such cases, the student is to learn the most expedient course to be taken among a natural array of possibilities. This is not true in the simple acquisition of knowledge—particularly verbal knowledge—where the task is only rarely to discriminate among responses in an array. In solving an equation, reporting a fact of history, restating the meaning of a sentence, or engaging in almost any of the other behavior which is the main concern of education, the student is to *generate* responses. He may generate and reject, but only rarely will he generate a set of responses from which he must then make a choice.

It may be argued that machines which provide for branching and decision-making are designed to teach more than verbal repertoires—in particular, that they will teach thinking. There are strategies in choosing from an array, for example, which require kinds of behavior beyond the mere emission of correct responses. We may agree to this without questioning the value of knowledge in the sense of a verbal repertoire. (The distinction is not between rote and insightful learning, for programmed instruction is especially free of rote memorizing in the etymological sense of wearing down a path through repetition.) If an "idea" or "proposition" is defined as something which can be expressed in many ways, then it may be taught by teaching many of these "ways." What is learned is more likely to generalize to comparable situations than a single syntactical form, and generalization is what distinguishes so-called deeper understanding.

But not all thinking is verbal. There are, first of all, alternative, parallel nonverbal repertoires. The mathematician begins with a verbal problem and ends with a verbal solution, but much of his intervening behavior may be of a different nature. The student who learns to follow or construct a proof entirely by manipulating symbols may not engage in this kind of thinking. Similarly, a merely verbal knowledge of physics, as often seen in the student who has "memorized the text," is

of little interest to the serious educator. Laboratories and demonstrations sometimes supply contingencies which build some nonverbal knowledge of physics. Special kinds of teaching machines could help, for machines are not only not confined to verbal instruction, they may well make it possible to reduce the emphasis on verbal communication between teacher and student.

A more clear-cut example of the distinction between verbal and nonverbal thinking is musical composition. The composer who "thinks musically" does more than perform on an instrument or enjoy music. He also does more than use musical notation. In some sense he "thinks" pitches, intervals, melodies, harmonic progressions, and so on. It should not surprise us that individuals differ greatly in their "abilities" to do this, since the necessary contingencies are in very short supply. One might attack the problem by setting up an explicit kinesthetic repertoire in which "thinking a pitch" takes the form of identifying a position on a keyboard. A device which arranges the necessary contingencies is under development. With its help we may discover the extent to which students can in general learn (and at what ages they can learn most effectively) to strike a key which produces a tone which has just been heard. Similar devices might generate important forms of nonverbal mathematical behavior or the behavior exhibited, say, by an inventor conceiving of a device in three dimensions, as well as creative repertoires in other forms of art. Here is an extraordinary challenge to the technology of instrumentation.

There is another sense in which the student must learn to think. Verbal and nonverbal repertoires may prepare him to behave in effective ways, but he will inevitably face novel situations in which he cannot at first respond appropriately. He may solve such problems, not by exercising some mental ability, but by altering either the external situation or the relative probabilities of parts of his own repertoire. In this way he may increase the probability of an adequate response.

In this sense, thinking consists of a special repertoire which we may call self-management. For example, the student may alter the extent to which the environment affects him by "attending" to it in different ways. As one step in teaching thinking we must teach effective attending. The phrase "Pay attention!" is as common on the lips of teachers as "Open, please" on those of dentists—and for much the same reason: both phrases set up working conditions. The student may pay

attention to avoid punishment and in doing so may learn to pay attention, but where aversive sanctions have been given up, teachers have resorted to attracting and holding attention. The techniques of the publication and entertainment industries are extensively invoked. Primers are usually decorated with colored pictures, and high school textbooks are sometimes designed to resemble picture magazines. Films dramatize subject matters in competition with noneducational films and television.

Attention which is captured by attractive stimuli must be distinguished from attention which is "paid." Only the latter must be learned. Looking and listening are forms of behavior, and they are strengthened by reinforcement. A pigeon can learn to match colors, for example, only if it "pays attention to them." The experimenter makes sure that it does so, not by attracting its attention, but by reinforcing it for looking. Similarly, a well-taught student pays attention to sentences, diagrams, samples of recorded speech and music, and so on, not because they are attractive but because something interesting occasionally happens *after* he has paid attention.

Most audio-visual devices fail to teach attention because they stimulate the student *before* he looks or listens closely. No matter how well a four-colored text or a dramatically filmed experiment in physics attracts attention, it prepares the student only for comics, advertising, picture magazines, television programs, and other material which is *interesting on its face*. What is wanted is an adult who, upon seeing a page of black-and-white text, will read it because it may *prove* interesting. Unfortunately, the techniques associated with captured and paid attention are incompatible. Whenever a teacher attracts the attention of a student, he deprives him of an opportunity to learn to pay attention. Teaching machines, with their control over the consequences of action, can make sure that paying attention will be effectively reinforced.

Another activity associated with thinking is studying—not merely looking at a text and reading it but looking and reading *for the sake of future action*. Suppose we show a child a picture and later, in the absence of the picture, reinforce him generously for correct answers to questions about it. If he has done nothing like this before, he will probably not be very successful. If we then show him another picture, he may begin to behave in a different way: he may engage in behavior which will increase the probability that he will later answer questions correctly. It will be to his advantage (and to ours

as educators) if this kind of behavior is taught rather than left to chance. We teach a student "how to study" when we teach him to take notes, to rehearse his own behavior, to test himself, to organize, outline, and analyze, to look for or construct mnemonic patterns, and so on. Some of these behaviors are obvious, but others are of more subtle dimensions and admittedly hard to teach. Machines have an advantage in maintaining the contingencies required for indirect or mediated reinforcement.

Other aspects of thinking, including the solution of personal problems, can also be analyzed and directly programed. This is not current practice, however. Students are most often "taught to think" simply by thrusting them into situations in which already established repertoires are inadequate. Some of them modify their behavior or the situation effectively and come up with solutions. They may have learned, but they have not necessarily been taught, how to think.

Logicians, mathematicians, and scientists have often tried to record and understand their own thinking processes, but we are still far from a satisfactory formulation of all relevant behaviors. Much remains to be learned about how a skillful thinker examines a situation, alters it, samples his own responses with respect to it, carries out specific verbal manipulations appropriate to it, and so on. It is quite possible that we cannot teach thinking adequately until all this has been analyzed. Once we have specified the behavior, however, we have no reason to suppose that it will then be any less adaptable to programed instruction than simple verbal repertoires.

Teaching machines and the associated practices of programed instruction will have proved too successful if their practical consequences are allowed to overshadow their promise for the future. We need teaching machines to help solve a very pressing problem, but we also need them to utilize our basic knowledge of human behavior in the design of entirely new educational practices.

Teaching machines are an example of the technological application of basic science. It is true that current machines might have been designed in the light of classroom experience and common sense, and that explanations of why they are effective can be paraphrased in traditional terms. The fact remains that more than half a century of the self-conscious examination of instructional processes had worked only moderate changes in educational practices. The laboratory study of learning provided the confidence, if not all the knowledge,

needed for a successful instrumental attack on the *status quo*. Traditional views may not have been actually wrong, but they were vague and were not entertained with sufficient commitment to work substantial technological changes.

As a technology, however, education is still immature, as we may see from the fact that it defines its goals in terms of traditional achievements. Teachers are usually concerned with reproducing the characteristics and achievements of already educated men. When the nature of the human organism is better understood, we may begin to consider not only what man has already shown himself to be, but what he may become under carefully designed conditions. The goal of education should be nothing short of the fullest possible development of the human organism. An experimental analysis of behavior, carried out under the advantageous conditions of the laboratory, will contribute to progress toward that goal. So will practical experiments conducted in schools and colleges with the help of adequate instrumentation.

Teaching Machines and Programed Instruction

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What are the advantages of programed instruction over the older audio-visual devices? How do teaching machines work? To what extent are teaching machines and programed instruction in use today? These are the questions to be discussed in this paper, for they frame a fourth query which is of great concern to education and the behavioral sciences today: What are the promises and prospects for teaching machines and programed instruction?

The older devices can also ask the student questions and provide some information that might help him answer. The teaching machine, however, probably does this more consistently than any other device. Typically, other devices ask questions only occasionally. And frequently they are asked only after an elaborate presentation of new information has been made—for example, at the end of a chapter.

Some of the other educational devices also tell the student that the response he made was correct. In a workbook, for example, the student can usually turn to a page at the back